

Name _____ Date _____ Period _____

Transformations of $f(x) = |x|$, Part I

Algebra II | Unit 2 | Day 2 | TEK A2.6.C

Objective: I can describe and graph the effect of a and b on $f(x) = a|bx|$ - vertical/horizontal stretches, compressions, and reflections.

Vocabulary

Transformation — a change to a parent function's graph: _____.

Vertical stretch/compression — $af(x)$; $|a| > 1$ _____ away from the x-axis, $0 < |a| < 1$ _____ toward it.

Horizontal stretch/compression — $f(bx)$; $|b| > 1$ _____ toward the y-axis, $0 < |b| < 1$ _____ away from it.

Reflection — $a < 0$ flips over the _____; $b < 0$ flips over the _____.

Part 1 — The Effect of a in $f(x) = a|x|$

Function	a-value	Effect	Vertex
$f(x) = x $	1	parent function	(0, 0)
$f(x) = 2 x $	2	_____	_____
$f(x) = (1/2) x $	1/2	_____	_____
$f(x) = - x $	-1	_____	_____

Notice: no matter what a is, the vertex never moves - $a(0)$ is always 0.

You Try. Describe $f(x) = 4|x|$: _____ Describe $f(x) = -(1/3)|x|$: _____

Part 2 — The Effect of b in $f(x) = |bx|$

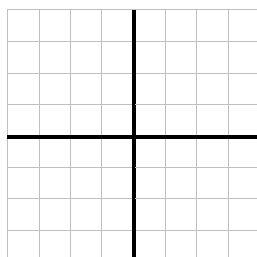
Function	b-value	Effect
$f(x) = x $	1	parent function
$f(x) = 2x $	2	_____
$f(x) = (1/2)x $	1/2	_____
$f(x) = -x $	-1	_____

You Try. Describe $f(x) = |3x|$: _____ Describe $f(x) = |(1/5)x|$: _____

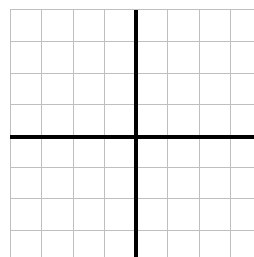
Part 3 — Graph and Compare

Graph both on their own grid. Each square is 1 unit; both axes run -5 to 5. Which one looks steeper?

A. $f(x) = 2|x|$



B. $f(x) = (1/2)|x|$



Steeper graph: _____

Part 4 — Rank Them

Order from FLATTEST to STEEPEST: $|x|$, $3|x|$, $(1/4)|x|$, $-2|x|$

Practice

1. Describe the transformation of $f(x) = 5|x|$ from the parent. _____
2. Describe the transformation of $f(x) = |(1/4)x|$ from the parent. _____
3. Describe the transformation of $f(x) = -6|x|$ from the parent.

4. Which of these two graphs is steeper: $f(x) = 3|x|$ or $f(x) = (1/5)|x|$?

5. $f(x) = -4|2x|$ combines an a-effect and a b-effect. Describe both.

6. True or False: $f(x) = |-5x|$ looks different from $f(x) = 5|x|$. Explain.

7. Describe the transformation of $f(x) = (1/3)|-x|$ from the parent.

8. Match each function to its effect: $f(x) = 7|x|$, $f(x) = |7x|$, $f(x) = -|x|$, $f(x) = (1/7)|x|$.
A. vertical compression B. reflection only C. vertical stretch by 7 D. horizontal compression
 $f(x) = 7|x|$: _____ $f(x) = |7x|$: _____ $f(x) = -|x|$: _____ $f(x) = (1/7)|x|$: _____
9. Two functions, $f(x) = a|x|$ and $g(x) = |bx|$, look IDENTICAL for every x you try. What must be true of a and b ?

10. Describe the transformation of $f(x) = -(2/3)|x|$, using both the stretch/compression amount and the reflection.

Exit Ticket

Describe every transformation in $f(x) = -3|x|$ from the parent function (stretch/compression amount, and any reflection).
